

Inequalities – 2

Hölder, Minkowski and the power means

Homework

10 сентября 2026 года

Theoretical minimum. Young's inequality and its convexity proof; Hölder and Minkowski in the discrete and integral forms and their equality cases; the power means M_p , monotonicity in p and the limiting values at 0 and $\pm\infty$; the weighted arithmetic–geometric mean inequality; log-convexity of $p \mapsto \log \|f\|_p$.

1. (Putnam 2021) Determine the maximum value of $S = \sum_{n=1}^{\infty} \frac{n}{2^n} (a_1 a_2 \cdots a_n)^{1/n}$ over all sequences of non-negative reals with $\sum a_k = 1$.
2. (Putnam 2005) Find all positive integers $n, k_1, \dots, k_n$ with $k_1 + \cdots + k_n = 5n - 4$ and $\frac{1}{k_1} + \cdots + \frac{1}{k_n} = 1$.
3. (IMO 2012) Let $n \geq 3$ and let $a_2, \dots, a_n$ be positive reals with $a_2 a_3 \cdots a_n = 1$.
 1. Prove that $(1 + a_2)^2 (1 + a_3)^3 \cdots (1 + a_n)^n > n^n$.
4. (IMO 2000) Let A, B, C be positive reals with product 1. Prove that $(A - 1 + \frac{1}{B})(B - 1 + \frac{1}{C})(C - 1 + \frac{1}{A}) \leq 1$.
5. (IMO 2008) Prove that $\frac{x^2}{(x-1)^2} + \frac{y^2}{(y-1)^2} + \frac{z^2}{(z-1)^2} \geq 1$ for all reals x, y, z different from 1 with $xyz = 1$, and show that equality holds for infinitely many triples of rationals.
6. (Proposed problems) Let $n \geq 3$. Prove that $\frac{n}{4}$ is the least positive constant k for which $a_1 + \cdots + a_n + \frac{k(a_1 - a_n)^2}{a_1 + a_n} \geq \sqrt{n(a_1^2 + \cdots + a_n^2)}$ holds whenever $a_1 \geq \cdots \geq a_n \geq 0$.
7. (Proposed problems) Let $a_1, a_2, a_3, a_4 \geq 0$. Prove that $a_1 + a_2 + 3a_3 + 3a_4 + \sqrt{(a_1 - a_3)^2 + (a_2 - a_4)^2} \geq \sum_{k=1}^4 \sqrt{a_k^2 + a_{k+1}^2}$, where $a_5 = a_1$; the natural tool here is Minkowski's inequality read as the triangle inequality in the plane.

8. (VJIMC 2019) Determine the largest constant $K \geq 0$ such that $\frac{a^a(b^2+c^2)}{(a^a-1)^2} + \frac{b^b(c^2+a^2)}{(b^b-1)^2} + \frac{c^c(a^2+b^2)}{(c^c-1)^2} \geq K \left(\frac{a+b+c}{abc-1} \right)^2$ holds for all positive reals a, b, c with $ab + bc + ca = abc$.
9. Prove the interpolation inequality $\|f\|_q \leq \|f\|_p^\theta \|f\|_r^{1-\theta}$ for $p < q < r$ with $\frac{1}{q} = \frac{\theta}{p} + \frac{1-\theta}{r}$, and deduce that $p \mapsto \log \|f\|_{1/p}$ is convex.
10. (VJIMC 2018) Let $x_1, \dots, x_n$ be positive reals with $|x_i - x_j| \leq 1$ for all pairs $i < j$. Prove that $\frac{x_1}{x_2} + \frac{x_2}{x_3} + \dots + \frac{x_{n-1}}{x_n} + \frac{x_n}{x_1} \geq \frac{x_2+1}{x_1+1} + \frac{x_3+1}{x_2+1} + \dots + \frac{x_{n-1}+1}{x_{n-1}+1} + \frac{x_1+1}{x_n+1}$.
11. (RMO 2021) Find all continuous $f: [0, 1] \rightarrow [0, \infty)$ such that $\int_0^1 f \cdot \int_0^1 f^2 \dots \int_0^1 f^{2020} = \left(\int_0^1 f^{2021} \right)^{1010}$; the point is that equality in the chain of power mean inequalities forces f to be constant.